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In [2]: import nltk
import re
import string
```

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In [3]: from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet
```

```
In [4]: def clean_text(text):
text = text.lower()

text = re.sub(r'https?//\S+', '', text)

text = re.sub(r'\S+@\S+', '', text)

text = re.sub(r'\d+', '', text)

text = re.sub(r'^\w\s', '', text)

text = re.sub(r'\s+', '', text).strip()
```

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In [6]: stop_words = set(stopwords.words('english'))
text = "The students are studying Artificial Intelligence"
words = word_tokenize(text.lower())

fitered = [w for w in words
            if w not in stop_words
            and w.isalpha()]
print("After stopwords remval:", fitered)

ps = PorterStemmer()
stemmed = [ps.stem(w) for w in fitered]
print("After stemming:", stemmed)
```

After stopwords remval: ['students', 'studying', 'artificial', 'intelligence']
 After stemming: ['student', 'studi', 'artifici', 'intellig']

```
In [8]: lemmatizer = WordNetLemmatizer()

print(lemmatizer.lemmatize('running'))
print(lemmatizer.lemmatize('running', pos='v'))
print(lemmatizer.lemmatize('better', pos='a'))
print(lemmatizer.lemmatize('mice', pos='n'))
print(lemmatizer.lemmatize('worst', pos='a'))
```

running
 run
 good
 mouse
 bad

In []: